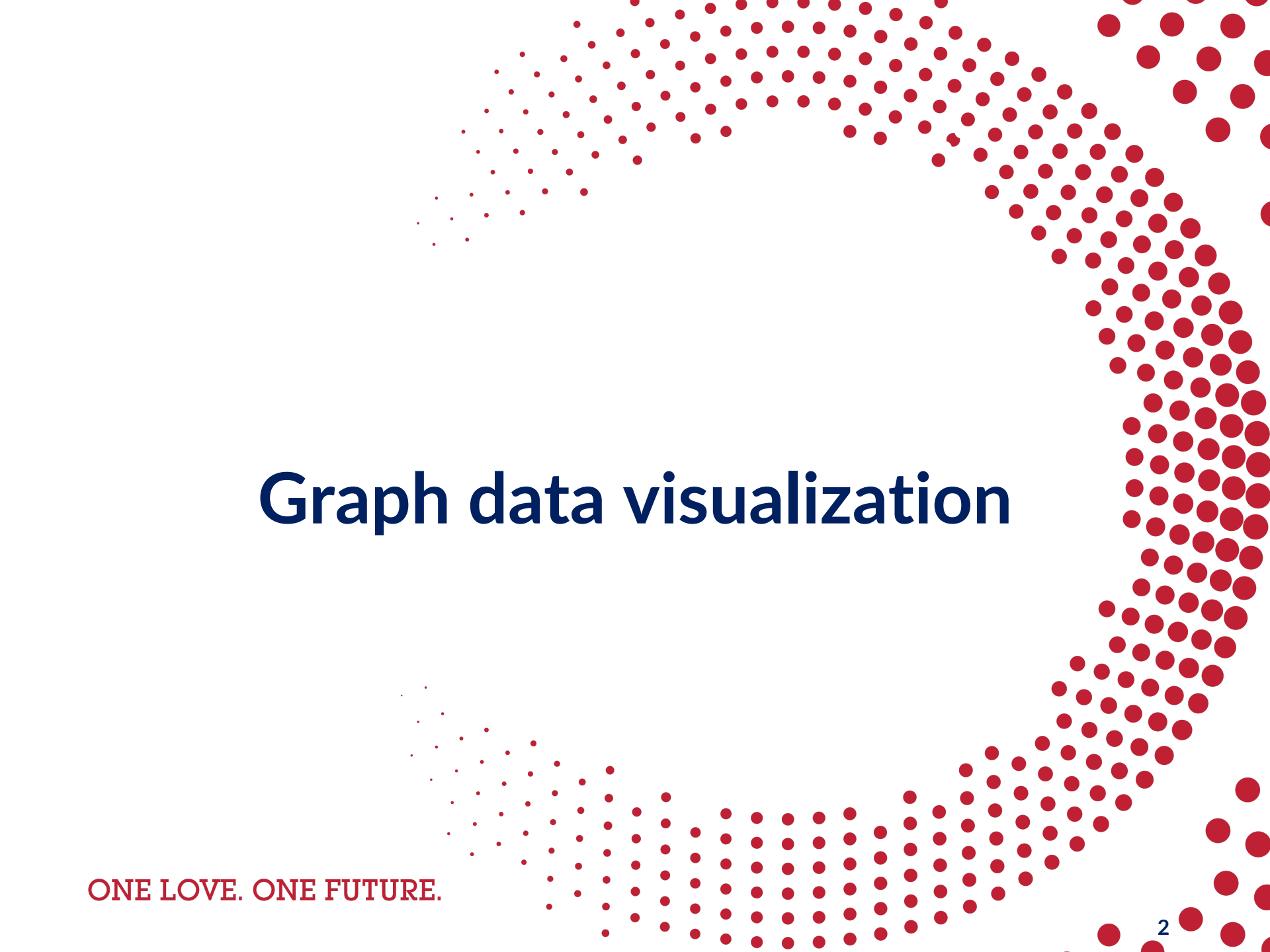




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ONE LOVE. ONE FUTURE.

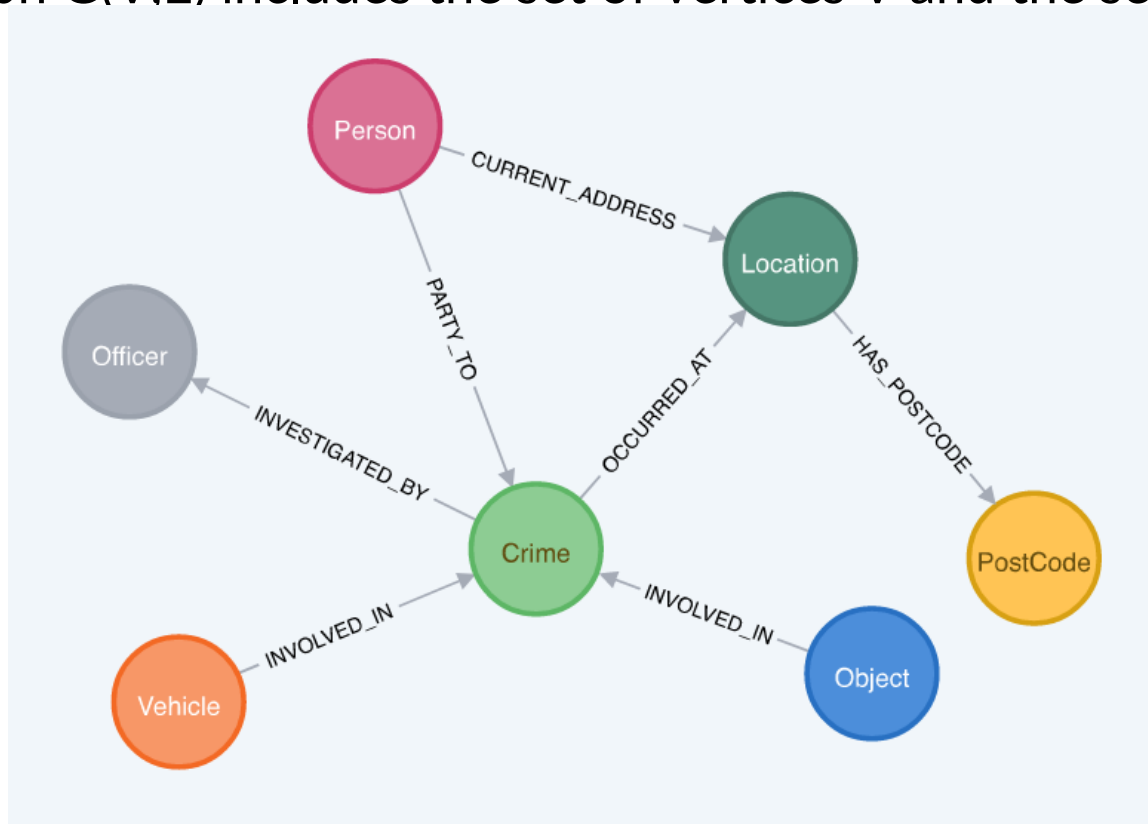
A decorative background consisting of numerous red dots of varying sizes arranged in a circular, spiral-like pattern that fills the right side and top of the slide.

Graph data visualization

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What is graph data

- Data representing relationships between entities
 - Entities are represented by nodes (vertices or points)
 - Relationships are represented by edges (lines or arcs)
 - The graph $G(V,E)$ includes the set of vertices V and the set of edges E

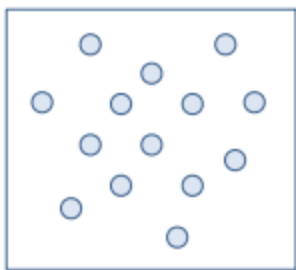


Properties of Graph Data

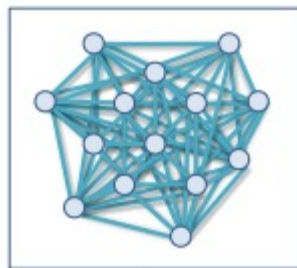
- Sparsity
 - Not all nodes are connected by edges
 - Number of edges is often much less than the number of possible connections between nodes
- Directionality
 - **Directed edges:** Show a one-way relationship between nodes (e.g., following on Twitter)
 - **Undirected edges:** Show a two-way relationship between nodes (e.g., friendship on Facebook)
- Weights
 - Weights represent the strength or importance of the relationship between nodes
 - Can be numerical values, distances, times, or other quantities

Basic concepts

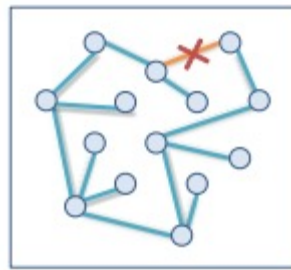
- An **independent set**, **stable set**, **coclique** or **anticlique** is a set of vertices in a graph, no two of which are adjacent.
- **Clique** is a subset of vertices of an undirected graph such that every two distinct vertices in the clique are adjacent.
- A **graph** is said to be **connected** if every pair of vertices in the graph is connected.
- A **tree** is an undirected graph in which any two vertices are connected by exactly one path, or equivalently a connected acyclic undirected graph.



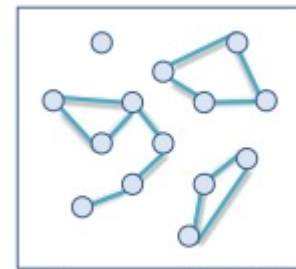
Independent Set



Clique



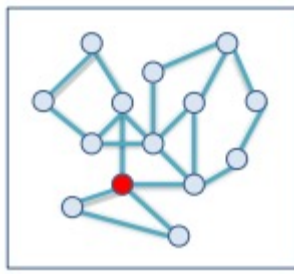
Tree



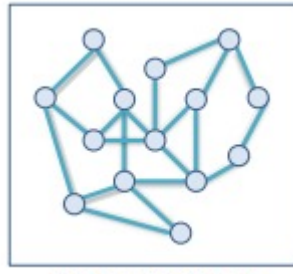
Unconnected Graph

Basic concepts

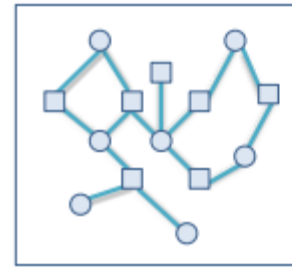
- An **articulation point (or cut vertex) (Điểm khớp)** is defined as a vertex which, when removed along with associated edges, makes the graph disconnected
- A **biconnected graph (đồ thị nối đôi)** is a connected and "nonseparable" graph, meaning that if any one vertex were to be removed, the graph will remain connected.
 - A biconnected graph has no articulation vertices.
- A **bipartite graph (or bigraph) (đồ thị hai phần)** is a graph whose vertices can be divided into two disjoint and independent sets U and V such that every edge connects a vertex in U to one in V



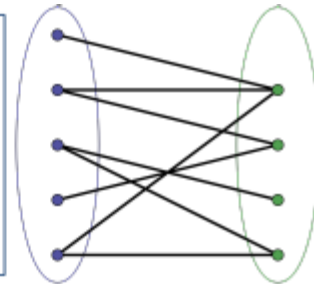
Articulation Point (red)



Biconnected Graph



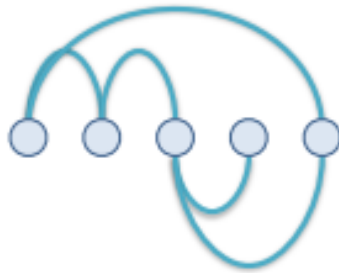
Bipartite Graph



Basic concepts

- Degree (bậc) of a vertex
 - $\deg(n)$ is the number of edges that are incident to the vertex
- Graph diameter (đường kính)
 - $\text{diam}(G)$ is the greatest distance (shortest path) between any pair of vertices

Graph visualization



Explicit
(Node-Link)

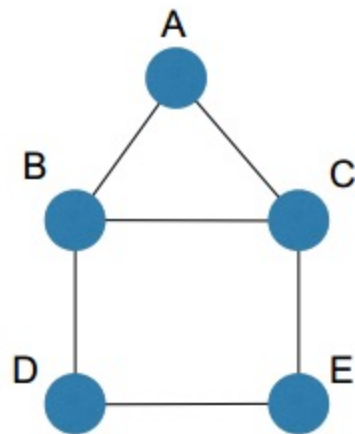


Matrix

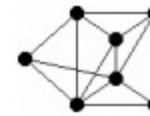


Implicit

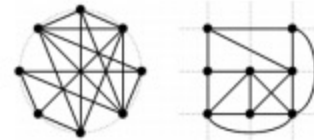
Explicit graph visualization (node-link)



Free



Styled



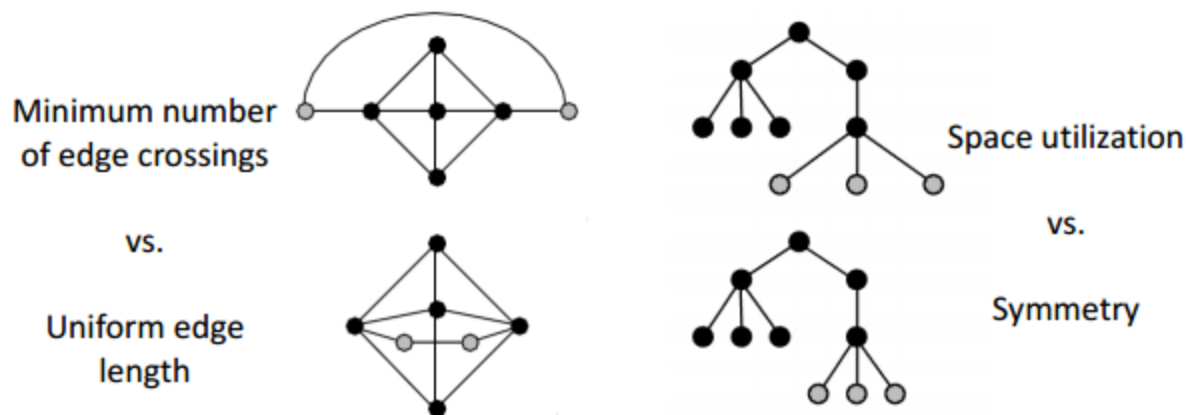
Fixed



H.J. Schulz 2006

Criteria for node-link representation

- Minimizing the intersection edges
- Minimize the distance between the vertices
- Minimize drawing area
- Edges of similar length
- Maximum angle between different edges
- Symmetry (graphs with the same structure must look the same)



Force-directed graph

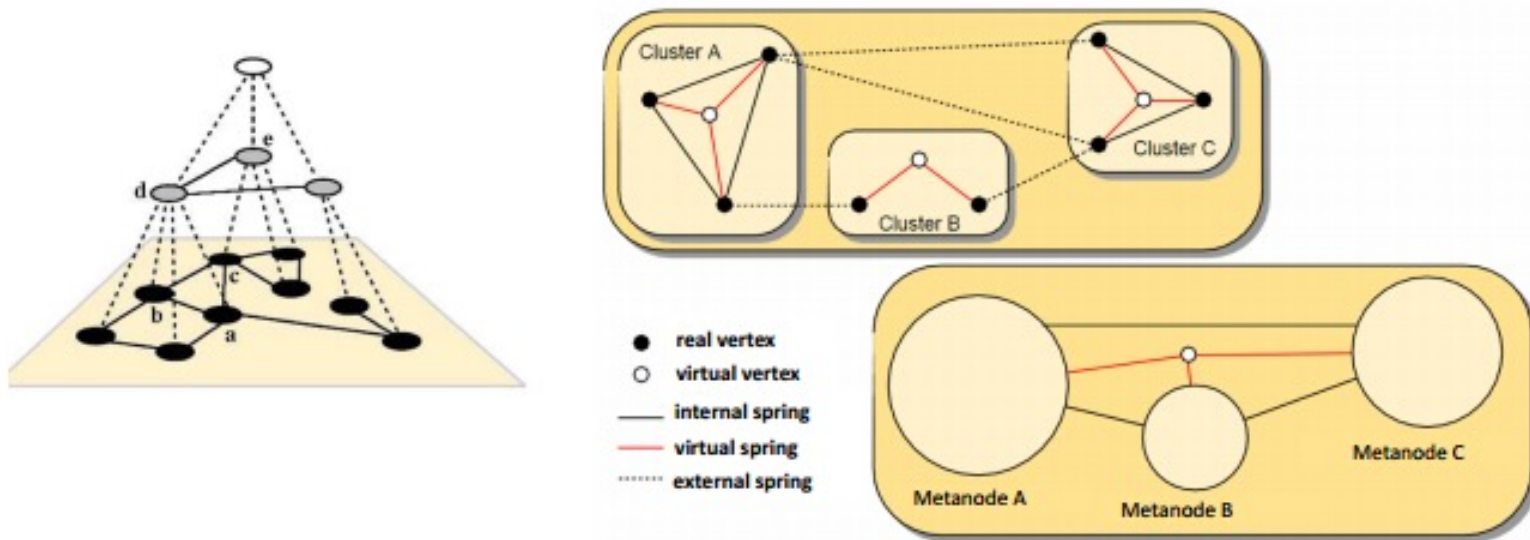
- The forces are applied to the nodes, pulling them closer together or pushing them further apart.
- This is repeated iteratively until the system comes to a mechanical equilibrium state



<https://observablehq.com/@d3/force-directed-graph>

Multi-level technique

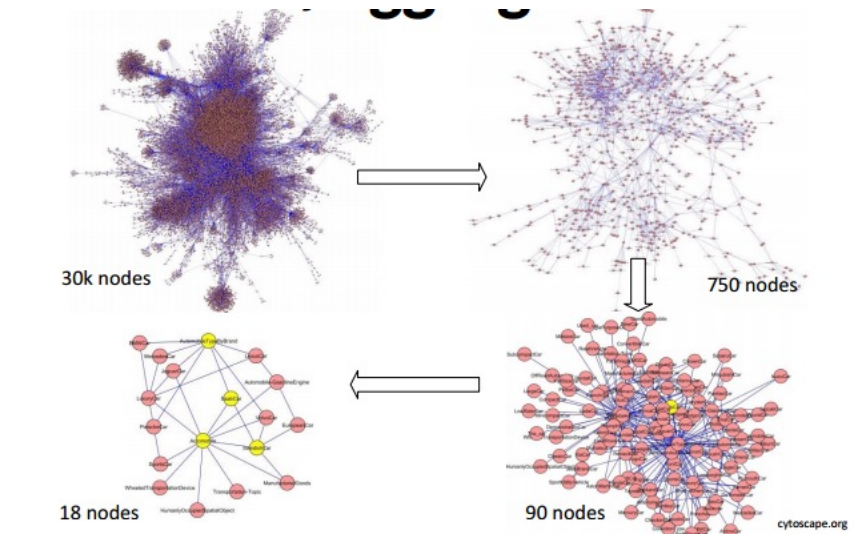
- Reduce the magnitude of a graph by merging vertices together, compute a partition on this reduced graph, and finally project this partition on the original graph.



[Schulz 2004]

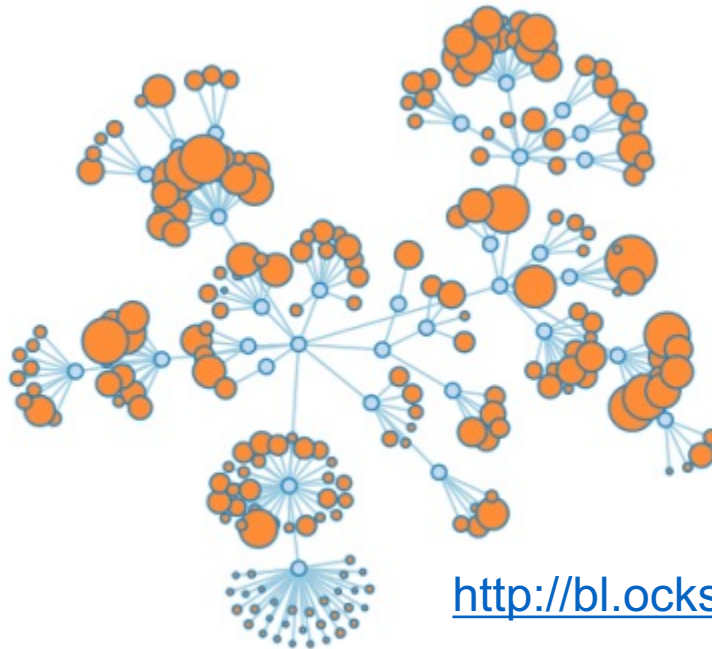
Sampling

- Large graphs can overwhelm visualization tools and viewers.
- Sampling allows us to explore a manageable subset of the data.
- Different sampling techniques highlight various aspects of the graph.
- Key sampling techniques
 - Random Sampling
 - Degree-Based Sampling
 - Ego-Network Sampling
 - Community Sampling



Collapse/Expand

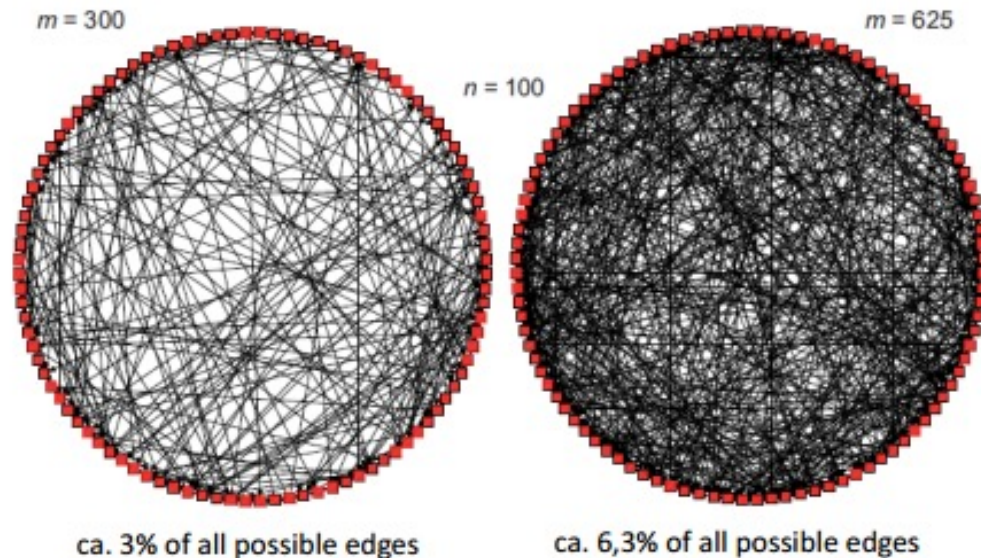
- Particularly useful for hierarchical graphs, where sub-trees can be collapsed or expanded as needed.
- **Clarity:** Reduces visual clutter, making it easier to focus on specific areas of interest.
- **Scalability:** Handles large datasets efficiently by allowing parts of the graph to be hidden.



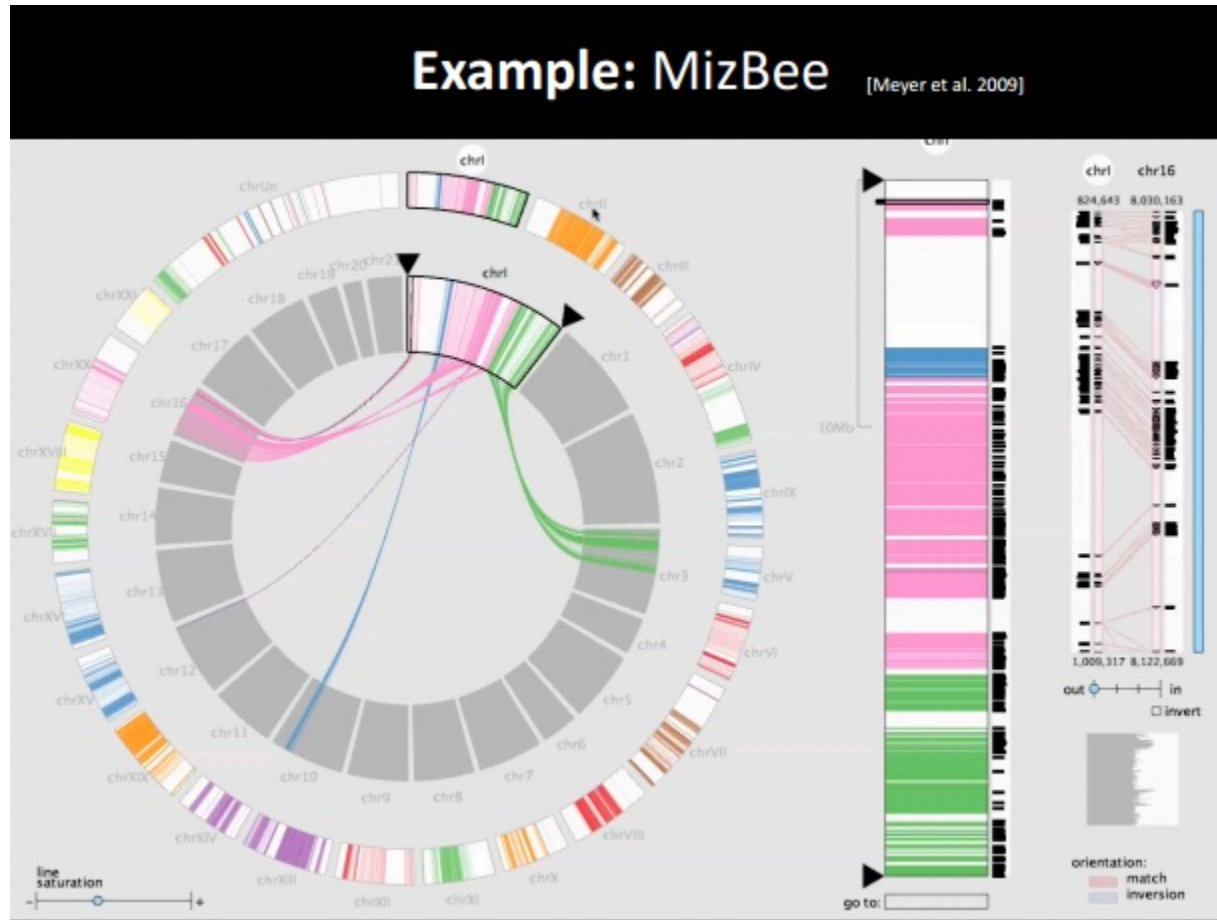
<http://bl.ocks.org/mbostock/1062288>

Fixed arrangement for graphs

- **Circos Layout**
 - **Compact Visualization:** Efficiently uses space to show large, complex datasets.
 - **Pattern Recognition:** Facilitates the identification of clusters, patterns, and relationships.
 - **Aesthetic Appeal:** Attractive and engaging, enhancing the visual storytelling of data.

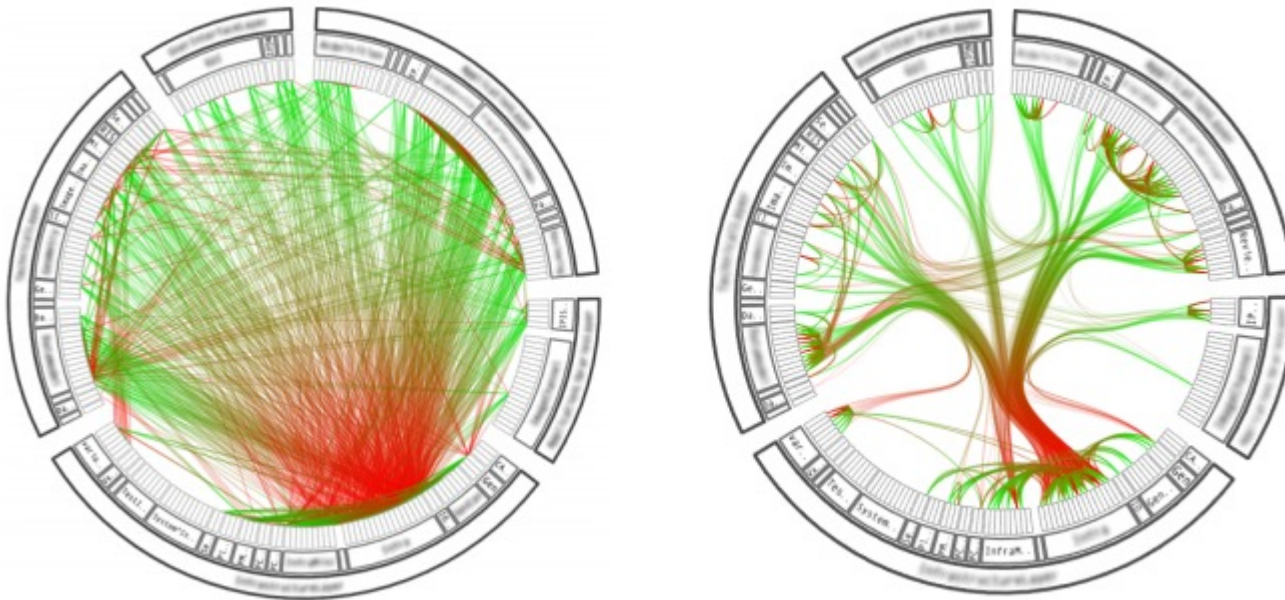


Fixed arrangement for graphs



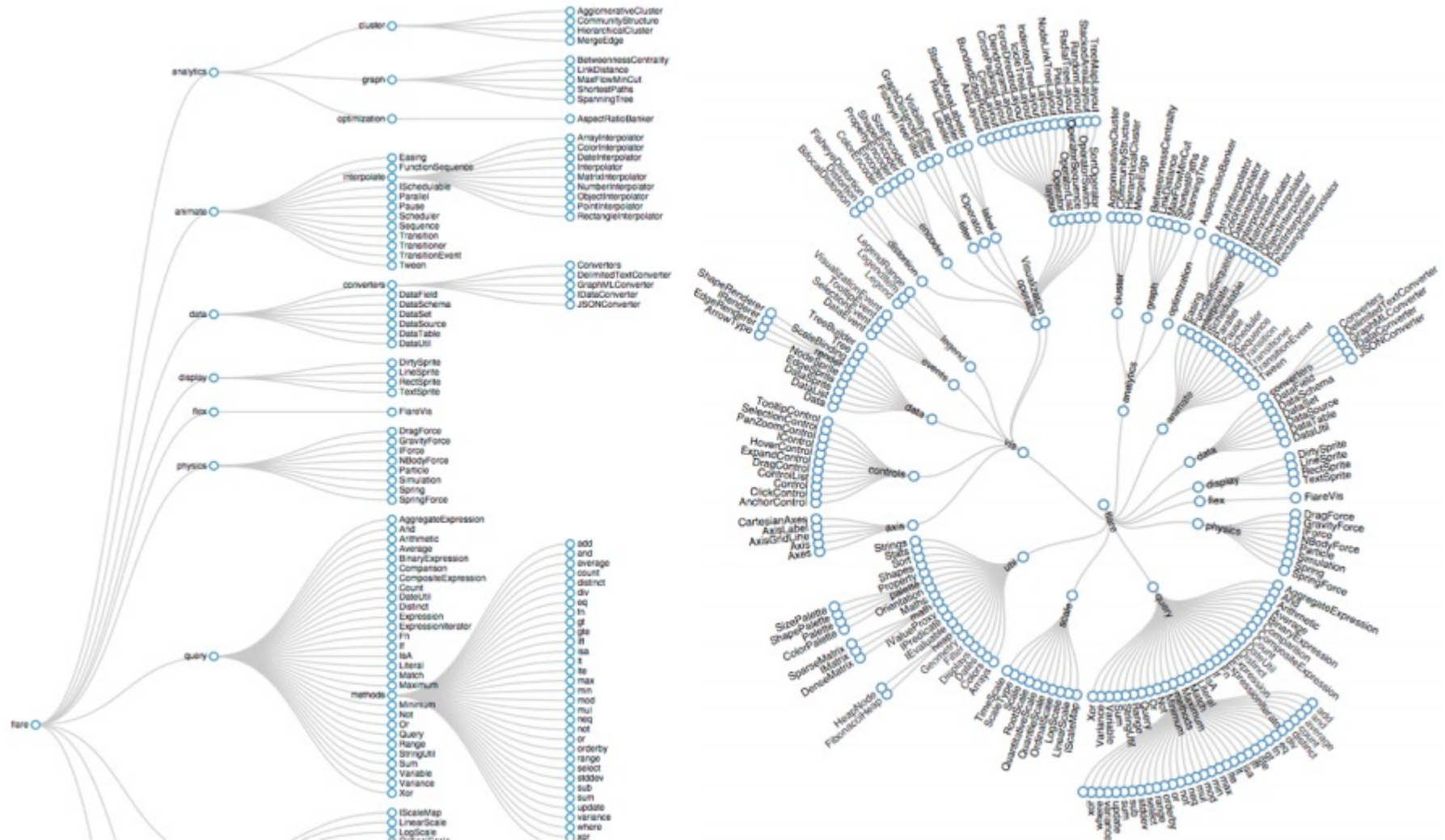
Fixed arrangement for graphs

- Edge bundling



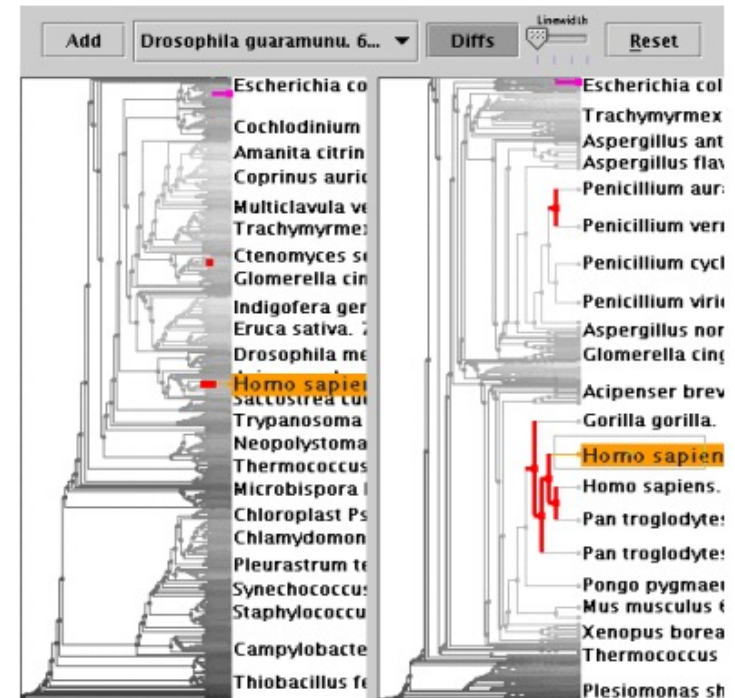
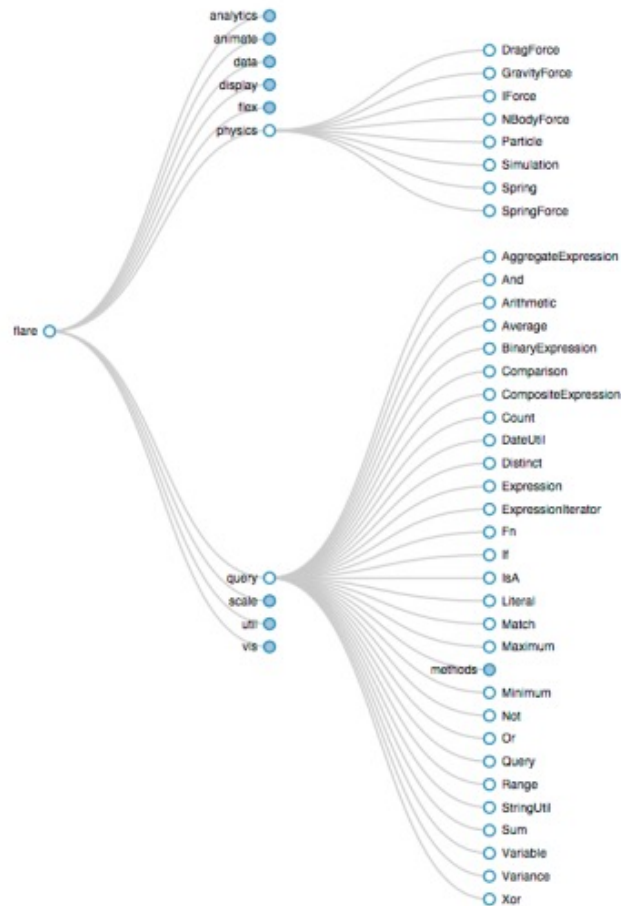
<https://observablehq.com/@d3/hierarchical-edge-bundling>

Tree layout



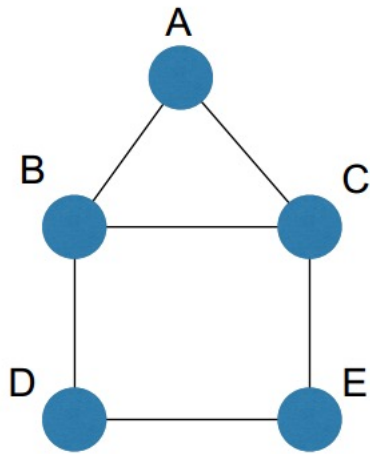
<https://observablehq.com/@d3/tree>

Tree layout



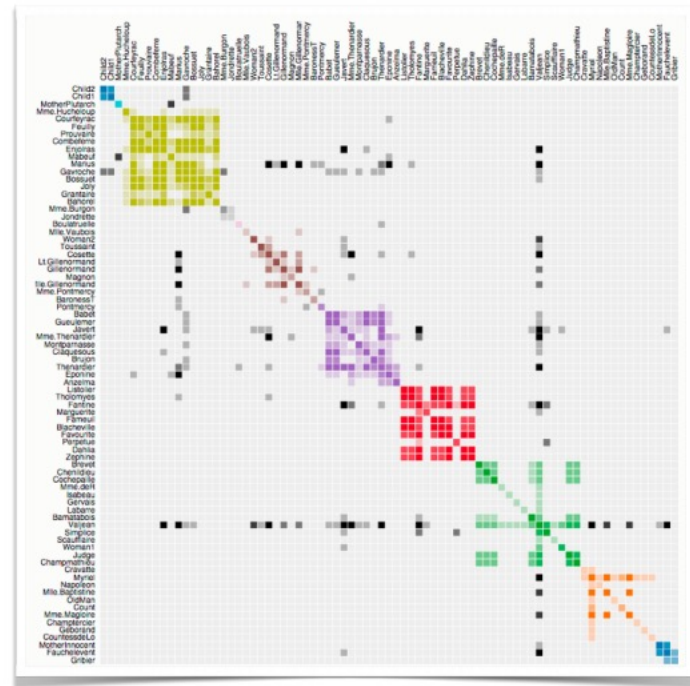
<http://bl.ocks.org/mbostock/4339083>

Adjacency matrix



	A	B	C	D	E
A					
B					
C					
D					
E					

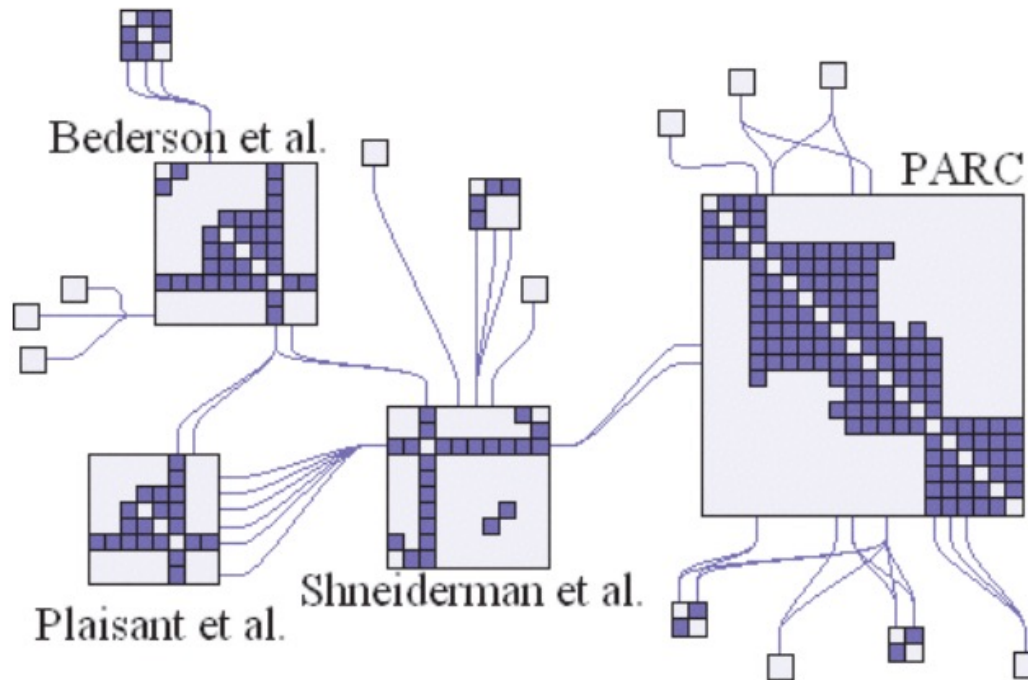
Adjacency matrix



Adjacency matrix

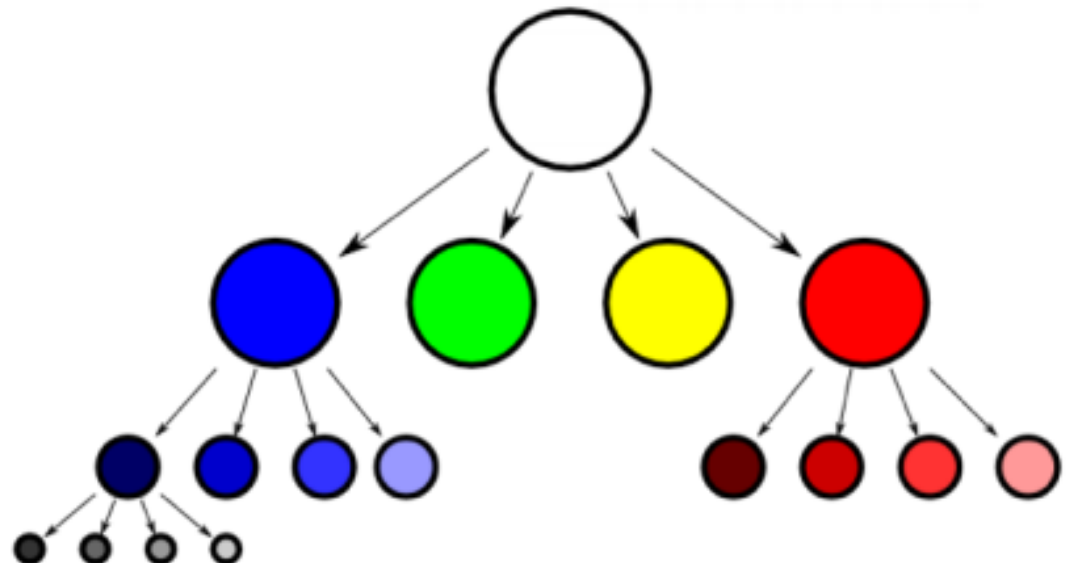
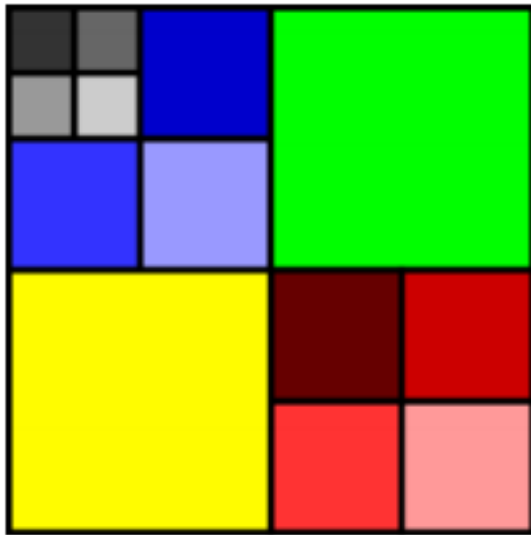
- Advantage
 - Can represent most graphs (except hypergraphs)
 - Focus on edges instead of vertices
 - No need to care about layout
- Disadvantage
 - Difficult to detect relationships such as paths, cycles, etc.

Hybrid layout



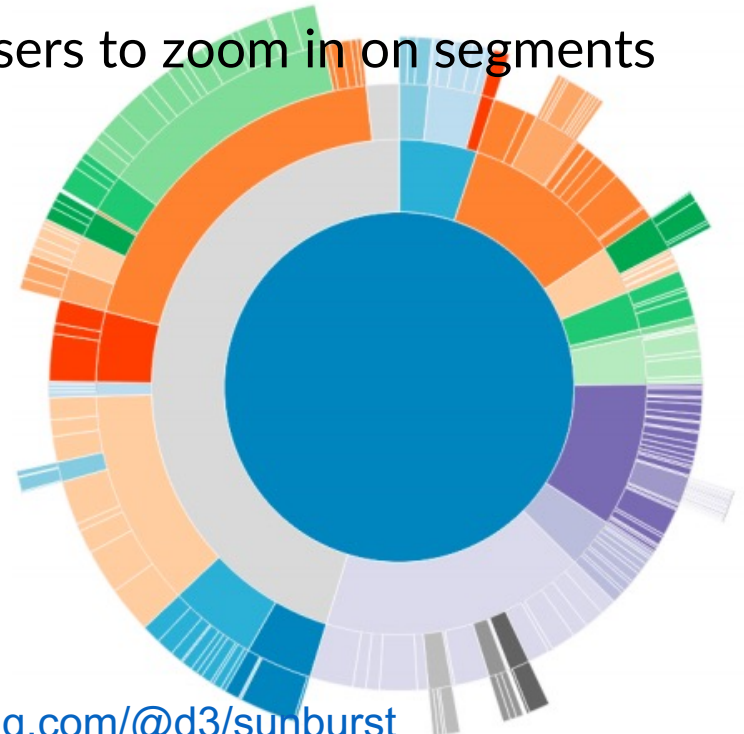
Implicit presentation

- A treemap chart is a visualization that displays hierarchical data using nested rectangles.
- Structure
 - The entire chart area represents the root of the hierarchy.
 - Each rectangle (or cell) represents a branch or leaf node.
 - Rectangles are sized and ordered by a quantitative variable.
 - Colors can be used to represent different categories or to encode additional data dimensions.



Implicit presentation

- A sunburst chart is a radial visualization that represents hierarchical data using concentric circles.
- Structure
 - The center circle represents the root of the hierarchy.
 - Each subsequent ring represents a level in the hierarchy.
 - Slices are divided into segments representing subcategories.
- Interactive Exploration: Often allows users to zoom in on segments for detailed views.



- <https://observablehq.com/@3cd7d5ec89dc7f68/graph-visualization-introduction>



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